

Homework #3 - MATH 257

Noah Arias

Problem 1

We have four I.I.D. random variables $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_4 \sim N_p(\mu, \Sigma)$

Part (a)

For $\mathbf{V}_1 = \frac{1}{4}\mathbf{X}_1 - \frac{1}{4}\mathbf{X}_2 + \frac{1}{4}\mathbf{X}_3 - \frac{1}{4}\mathbf{X}_4$, since $\mathbf{V}_1$ is a linear combination of I.I.D. normally distributed random variables, then V_1 is also normally distributed.

We rewrite the linear combination as a matrix product as follows:

$$\frac{1}{4}\mathbf{X}_1 - \frac{1}{4}\mathbf{X}_2 + \frac{1}{4}\mathbf{X}_3 - \frac{1}{4}\mathbf{X}_4 = \left(\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}\right) \cdot \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \mathbf{X}_3 \\ \mathbf{X}_4 \end{pmatrix}$$

Where $\mathbf{A} = \left(\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}\right)$ and $\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \mathbf{X}_3 \\ \mathbf{X}_4 \end{pmatrix}$.

In doing so, we can leverage the following result: $\mathbf{AX} \sim N_{V_1}(\mathbf{A}\mu, \mathbf{A}\Sigma\mathbf{A}^\top)$

The mean of $\mathbf{V}_1$ is the following:

$$E(\mathbf{V}_1) = E(\mathbf{AX}) = \mathbf{A}E(\mathbf{X}) = \left(\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}\right) \cdot \begin{pmatrix} \mu \\ \mu \\ \mu \\ \mu \end{pmatrix} = \frac{1}{4}\mu - \frac{1}{4}\mu + \frac{1}{4}\mu - \frac{1}{4}\mu = 0$$

The covariance matrix of

$$\Sigma_{\mathbf{V}_1} = \mathbf{A}\Sigma\mathbf{A}^\top$$

$$\begin{aligned}
&= \left(\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}\right) \cdot \begin{pmatrix} \sigma_{1,1} & \sigma_{1,2} & \sigma_{1,3} & \sigma_{1,4} \\ \sigma_{2,1} & \sigma_{2,2} & \sigma_{2,3} & \sigma_{2,4} \\ \sigma_{3,1} & \sigma_{3,2} & \sigma_{3,3} & \sigma_{3,4} \\ \sigma_{4,1} & \sigma_{4,2} & \sigma_{4,3} & \sigma_{4,4} \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{4} \\ -\frac{1}{4} \\ \frac{1}{4} \\ -\frac{1}{4} \end{pmatrix} \\
&= \left(\frac{1}{4}(\sigma_{1,1} - \sigma_{2,1} + \sigma_{3,1} - \sigma_{4,1}), \frac{1}{4}(\sigma_{1,2} - \sigma_{2,2} + \sigma_{3,2} - \sigma_{4,2}), \frac{1}{4}(\sigma_{1,3} - \sigma_{2,3} + \sigma_{3,3} - \sigma_{4,3}), \frac{1}{4}(\sigma_{1,4} - \sigma_{2,4} + \sigma_{3,4} - \sigma_{4,4})\right) \cdot \begin{pmatrix} \frac{1}{4} \\ -\frac{1}{4} \\ \frac{1}{4} \\ -\frac{1}{4} \end{pmatrix} \\
&= \frac{1}{16}(\sigma_{1,1} - \sigma_{2,1} + \sigma_{3,1} - \sigma_{4,1}) - \frac{1}{16}(\sigma_{1,2} - \sigma_{2,2} + \sigma_{3,2} - \sigma_{4,2}) \\
&+ \frac{1}{16}(\sigma_{1,3} - \sigma_{2,3} + \sigma_{3,3} - \sigma_{4,3}) - \frac{1}{16}(\sigma_{1,4} - \sigma_{2,4} + \sigma_{3,4} - \sigma_{4,4}) \\
&= \frac{1}{16}(\sigma_{1,1} - \sigma_{2,1} + \sigma_{3,1} - \sigma_{4,1} - \sigma_{1,2} + \sigma_{2,2} - \sigma_{3,2} + \sigma_{4,2} \\
&+ \sigma_{1,3} - \sigma_{2,3} + \sigma_{3,3} - \sigma_{4,3} - \sigma_{1,4} + \sigma_{2,4} - \sigma_{3,4} + \sigma_{4,4}) \\
&= \frac{1}{16}(\sigma_{1,1} - 2\sigma_{2,1} + 2\sigma_{3,1} - 2\sigma_{4,1} + \sigma_{2,2} - 2\sigma_{3,2} + 2\sigma_{4,2} + \sigma_{3,3} - 2\sigma_{4,3} + \sigma_{4,4})
\end{aligned}$$

Therefore, $\mathbf{V}_1 \sim N(0, \frac{1}{16}(\sigma_{1,1} - 2\sigma_{2,1} + 2\sigma_{3,1} - 2\sigma_{4,1} + \sigma_{2,2} - 2\sigma_{3,2} + 2\sigma_{4,2} + \sigma_{3,3} - 2\sigma_{4,3} + \sigma_{4,4}))$

Part (b)

Problem 2

Part (a)

Part (b)

Part (c)

Part (d)